

Q1 (20 July 2021 Shift 2)

A body at rest is moved along a horizontal straight line by a machine delivering a constant power. The distance moved by the body in time ' t ' is proportional to:

(1) $t^{\frac{3}{2}}$

(2) $t^{\frac{1}{2}}$

(3) $t^{\frac{1}{4}}$

(4) $t^{\frac{3}{4}}$

Q2 (25 July 2021 Shift 1)

Water droplets are coming from an open tap at particular rate. The spacing between a droplet observed at 4th second after its fall to the next

droplet is 34.3 m. At what rate the droplets are coming from the tap ? (Take $g = 9.8 \text{ m/s}^2$)

(1) 3 drops / 2 seconds

(2) 2 drops / second

(3) 1 drop / second

(4) 1 drop / 7 seconds

Q3 (25 July 2021 Shift 2)

The relation between time t and distance x for a moving body is given as $t = mx^2 + nx$, where m and n are constants. The retardation of the motion

(1) $2mv^3$

(2) $2mnv^3$

(3) $2nv^3$

(4) $2n^2v^3$

Q4 (25 July 2021 Shift 2)

Questions with Answer Keys

MathonGo

A balloon was moving upwards with a uniform velocity of 10 m/s. An object of finite mass is dropped from the balloon when it was at a height of 75 m from the ground level. The height of the balloon from the ground when object strikes the ground was around
(takes the value of g as 10 m/s^2)

- (1) 300 m
- (2) 200 m
- (3) 125 m
- (4) 250 m

Q5 (25 July 2021 Shift 2)

The instantaneous velocity of a particle moving in a straight line is given as $v = \alpha t + \beta t^2$, where α and β are constants. The distance travelled by the particle between 1 s and 2 s is :

- (1) $3\alpha + 7\beta$
- (2) $\frac{3}{2}\alpha + \frac{7}{3}\beta$
- (3) $\frac{\alpha}{2} + \frac{\beta}{3}$
- (4) $\frac{3}{2}\alpha + \frac{7}{2}\beta$

Q6 (27 July 2021 Shift 1)

A ball is thrown up with a certain velocity so that it reaches a height 'h'. Find the ratio of the two different times of the ball reaching $\frac{h}{3}$ in both the directions.

- (1) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$
- (2) $\frac{1}{3}$
- (3) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$
- (4) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

Q7 (27 July 2021 Shift 2)

Questions with Answer Keys

MathonGo

A particle of mass M originally at rest is subjected

to a force whose direction is constant but magnitude varies with time according to the relation

$$F = F_0 \left[1 - \left(\frac{t-T}{T} \right)^2 \right]$$

Where F_0 and T are constants. The force acts only

for the time interval $2T$. The velocity v of the particle after time $2T$ is :

(1) $2 F_0 T/M$

(2) $F_0 T/2M$

(3) $4 F_0 T/3M$

(4) $F_0 T/3M$

Answer Key**Q1 (1)****Q2 (3)****Q3 (1)****Q4 (3)****Q5 (2)****Q6 (3)****Q7 (3)**

Hints and Solutions

MathonGo

Q1

$$P = \text{constant}$$

$$\frac{1}{2}mv^2 = Pt$$

$$\Rightarrow v \propto \sqrt{t}$$

$$\frac{dx}{dt} = C\sqrt{t} \quad C = \text{constant}$$

by integration.

$$x \propto t^{3/2}$$

Q2

$$\text{In } 4\text{sec. } 1^{\text{st}} \text{ drop will travel } \Rightarrow \frac{1}{2} \times (9.8) \times (4)^2 = 78.4 \text{ m}$$

$$\therefore 2^{\text{nd}} \text{ drop would have travelled } \Rightarrow 78.4 - 34.3 = 44.1 \text{ m}$$

$$\text{Time for } 2^{\text{nd}} \text{ drop } \frac{1}{2}(9.8)t^2 = 44.1$$

$$t = 3 \text{ sec}$$

each drop have time gap of 1sec

$\therefore$ 1 drop per sec

Q3

$$t = mx^2 + nx$$

$$\frac{1}{v} = \frac{dt}{dx} = 2mx + n$$

$$v = \frac{1}{2mx+n}$$

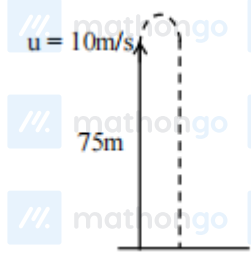
$$\frac{dv}{dt} = -\frac{2m}{(2mx+n)^2} \left(\frac{dx}{dt} \right)$$

$$a = -(2m)v^3$$

Q4

Hints and Solutions

MathonGo



Object is projected as shown so as per motion under gravity $S = ut + \frac{1}{2}at^2$

$$-75 = +10t + \frac{1}{2}(-10)t^2 \Rightarrow t = 5 \text{ sec}$$

Object takes $t = 5 \text{ s}$ to fall on ground Height of balloon from ground $H = 75 + ut$

$$= 75 + 10 \times 5 = 125 \text{ m}$$

Q5

$$V = \alpha t + \beta t^2$$

$$\frac{ds}{dt} = \alpha t + \beta t^2$$

$$\int_{S_1}^{S_2} ds = \int_1^2 (\alpha t + \beta t^2) dt$$

$$S_2 - S_1 = \left[\frac{\alpha t^2}{2} + \frac{\beta t^3}{3} \right]_1^2$$

As particle is not changing direction So distance = displacement. Distance = $\left[\frac{\alpha[4-1]}{2} + \frac{\beta[8-1]}{3} \right]$

$$= \frac{3\alpha}{2} + \frac{7\beta}{3}$$

Q6

$$u = \sqrt{2gh}$$

$$\text{Now, } S = \frac{h}{3} \quad a = -g$$

$$S = ut + \frac{1}{2}at^2$$

$$\frac{h}{3} = \sqrt{2gh}t + \frac{1}{2}(-g)t^2$$

$$t^2 \left(\frac{g}{2} \right) - \sqrt{2gh}t + \frac{h}{3} = 0$$

From quadratic equation

$$t_1, t_2 = \frac{\sqrt{2gh} \pm \sqrt{2gh - \frac{4g}{2} \frac{h}{3}}}{g}$$

Hints and Solutions

MathonGo

$$\frac{t_1}{t_2} = \frac{\sqrt{2gh} - \sqrt{\frac{4gh}{3}}}{\sqrt{2gh} + \sqrt{\frac{4gh}{3}}}$$

$$= \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$$

Q7

$$t = 0, u = 0$$

$$a = \frac{F_0}{M} - \frac{F_0}{MT^2} (t - T)^2 = \frac{dv}{dt}$$

$$\int_0^v dv = \int_{t=0}^{2T} \left(\frac{F_0}{M} - \frac{F_0}{MT^2} (t - T)^2 \right) dt$$

$$V = \left[\frac{F_0}{M} t \right]_0^{2T} - \frac{F_0}{MT^2} \left[\frac{t^3}{3} - t^2 T + T^2 t \right]_0^{2T}$$

$$V = \frac{4 F_0 T}{3M}$$